Neural Face Identifier – Project Proposal

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`github.com/richardkocian/neural-face-identifier`

1 Problem Description

This project develops a fine-tuned deep learning model for face identification in historical scans. Unlike standard datasets, these documents face challenges like low resolution, halftone artifacts, and archival noise. We aim to bridge the domain gap between modern photography and historical print for reliable identification in digital libraries.

2 Related Work & Inspiration

Vision Transformer (ViT-S/8): We utilize the ViT-S/8 architecture, which processes images as sequences of 8×8 pixel patches. We use a backbone pre-trained on the MS1MV3 dataset to leverage representations learned from millions of identities.
Model: `https://huggingface.co/gaunernst/vit_small_patch_gap_112.cosface_ms1mv3`
CosFace (Large Margin Cosine Loss): For fine-tuning, we use CosFace, which normalizes features and weights onto a hypersphere, introducing a cosine margin (m) in the angular space. This explicitly minimizes intra-class variance and maximizes inter-class variance, significantly improving discriminative power in noisy domains.
Paper: `https://arxiv.org/abs/1801.09414`

3 Datasets

- **MS1MV3:** A cleaned version of MS-Celeb-1M, serving as the source for our pre-trained backbone.
- **People-Gator:** Dataset of face detections from scanned historical magazines provided by the mentor.
- **WikiFace:** An additional benchmark to test generalization, also provided by the mentor.

Dataset	Identities	Images	Resolution	Role	Size
MS1MV3	93,431	5,179,510	112x112 px	Pre-training	29 GB
People-Gator	589	30,700	See Fig. 1	Fine-tuning & Evaluation	6.62 GB
WikiFace	1,537	3,223	112x112 px	Evaluation	10.1 MB

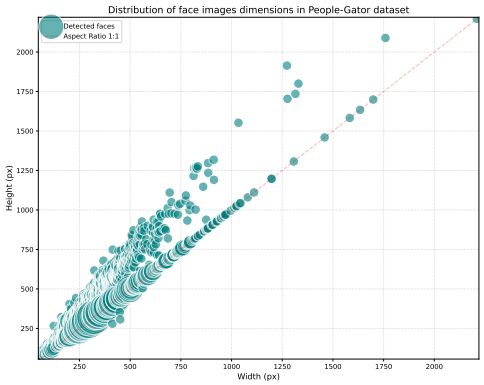

Figure 1: People-Gator Resolution Dist.

Metric	Width (px)	Height (px)
Mean / Median	351.67 / 318.00	412.00 / 370.00
Std. Deviation	161.21	183.43

Table 1: Detailed Statistics for People-Gator.

4 Proposed Solution & Plan

We adopt a transfer learning strategy to adapt a high-performance transformer to scanned documents:

1. **Data Pipeline:** Transformation and preprocessing of "People-Gator" for the PyTorch training loop.
2. **Backbone:** Selection of a pre-trained ViT-S/8 (GAP) architecture.
3. **Fine-tuning:** Training with CosFace loss; experimenting with freezing strategies to prevent forgetting.
4. **Domain Adaptation:** (Optional) Applying synthetic degradations (noise, blurring, halftone filters) to subset of MS1MV3 dataset to have more data for fine-tuning.

4.1 Experiments & Evaluation Metrics

We will quantitatively evaluate the impact of fine-tuning across the following scenarios:

Experiment	Model	Training Data	Evaluation Dataset
Baseline 1	ViT-S/8 (Pre-trained)	MS1MV3	People-Gator (Test)
Baseline 2	ViT-S/8 (Pre-trained)	MS1MV3	WikiFace
Finetuned 1	ViT-S/8 (Ours)	People-Gator	People-Gator (Test)
Finetuned 2	ViT-S/8 (Ours)	People-Gator	WikiFace

Table 2: Experiments Summary